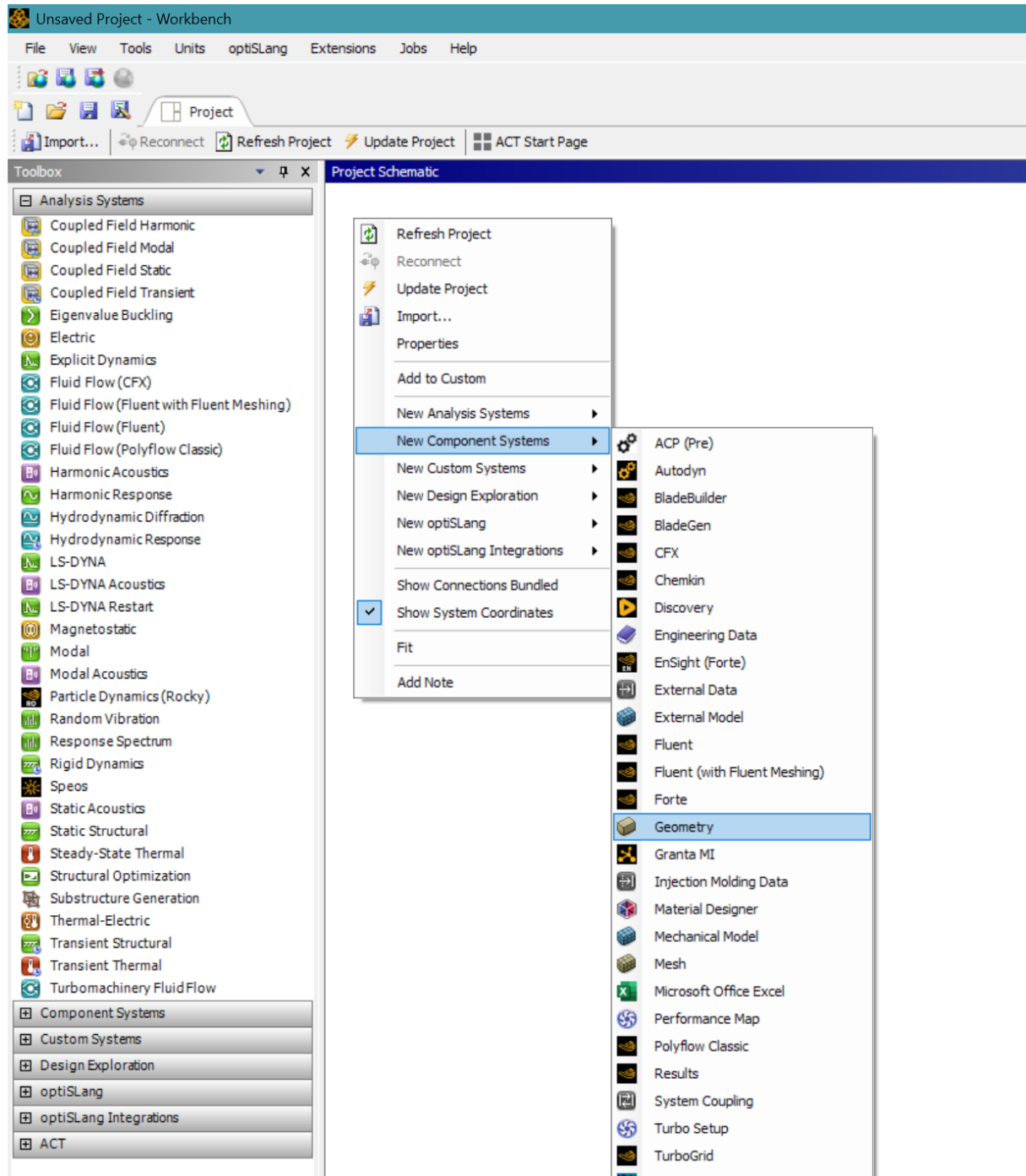


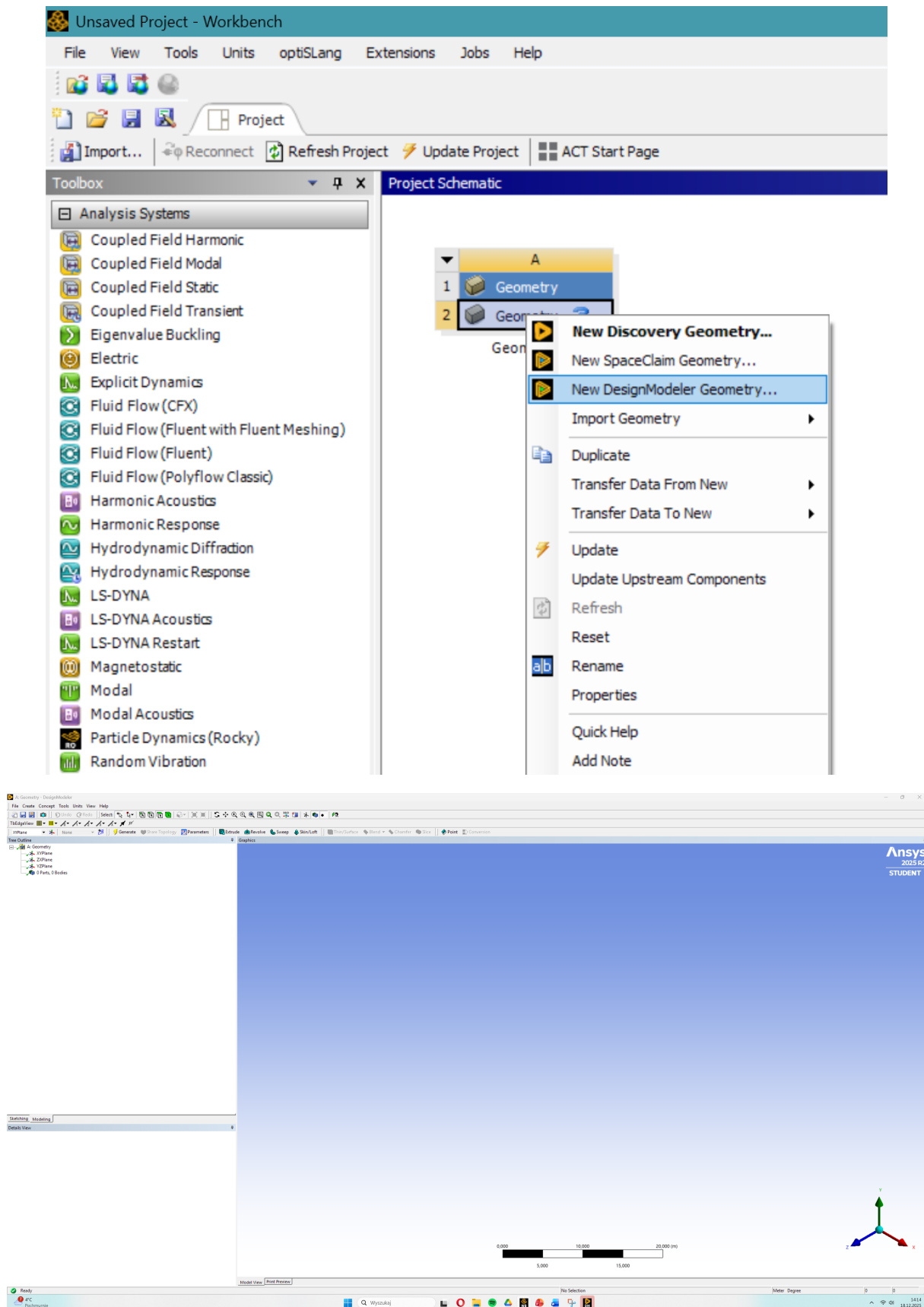
ANSYS

Otwieramy **Workbench**.

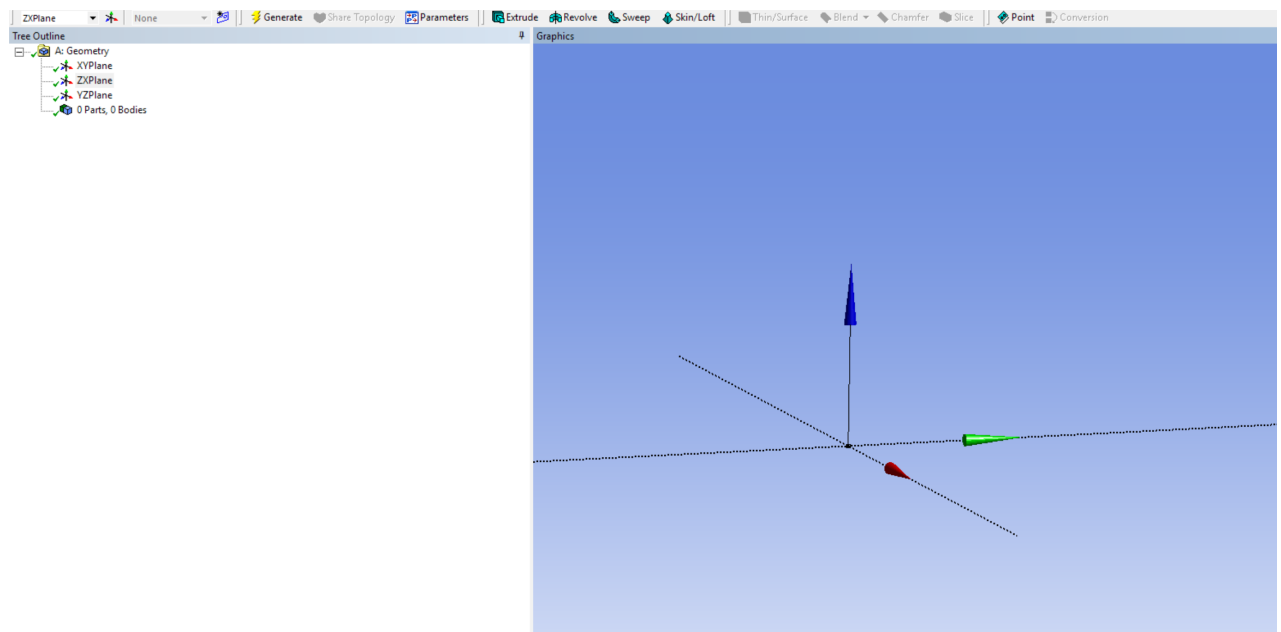
Prawy przycisk myszy → **New Component System** → **Geometry**.



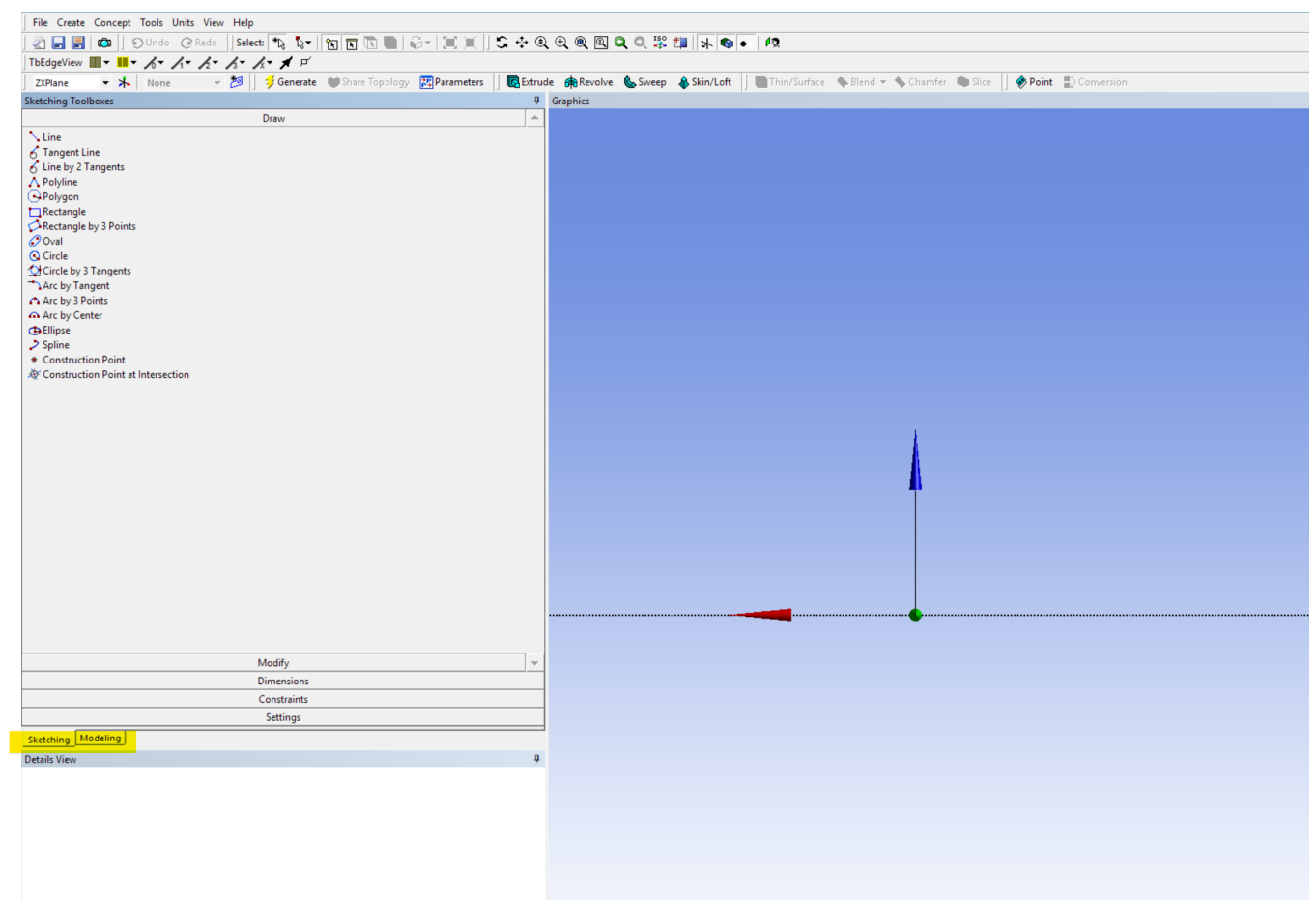
Prawy przycisk myszy na **Geometry** → **New DesignModeler**



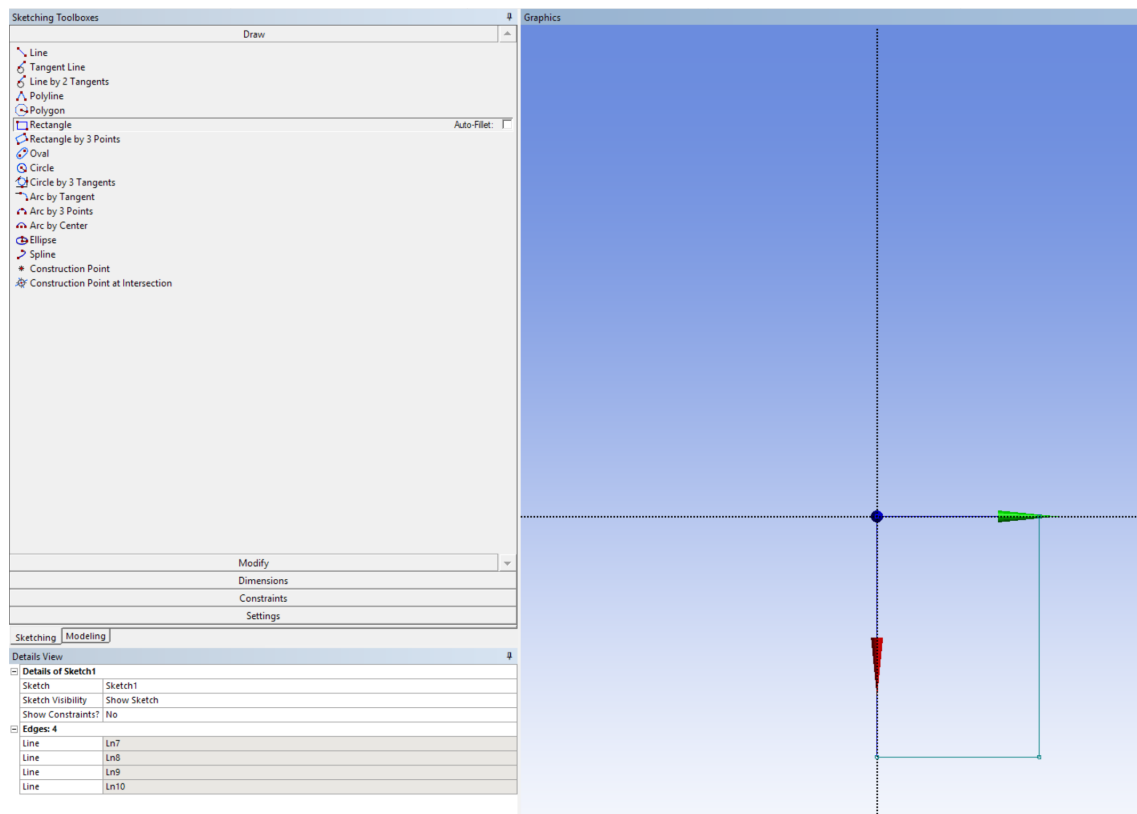
Klikamy **ZXPlane**.



Przechodzimy do zakładki **Sketching**.



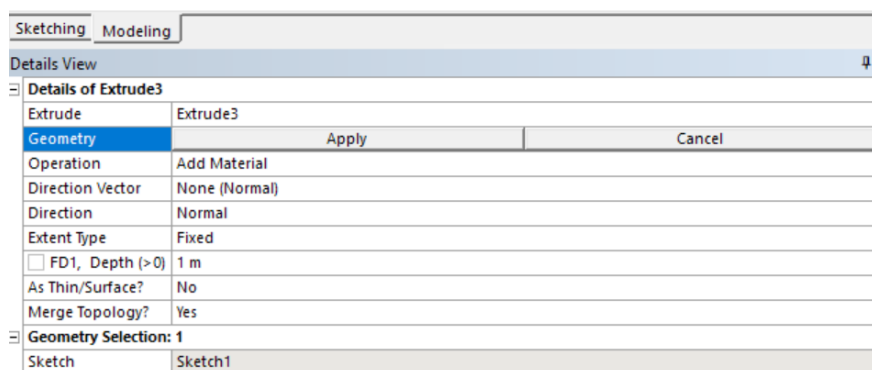
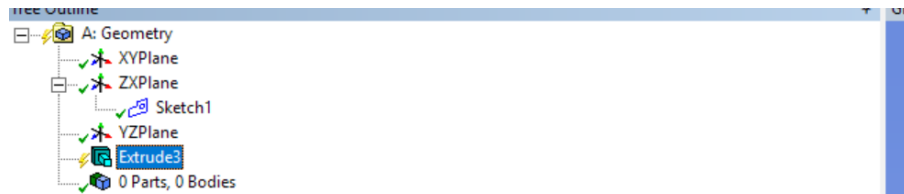
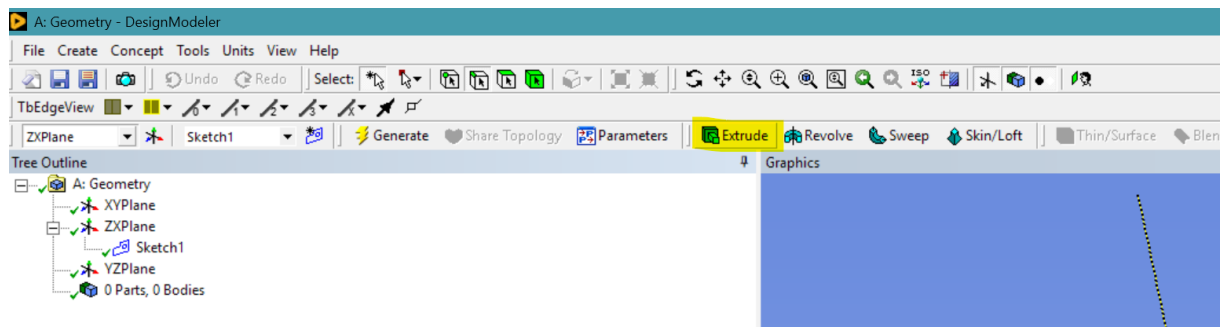
Rysujemy to, co chcemy (ja narysowałam prostokąt).



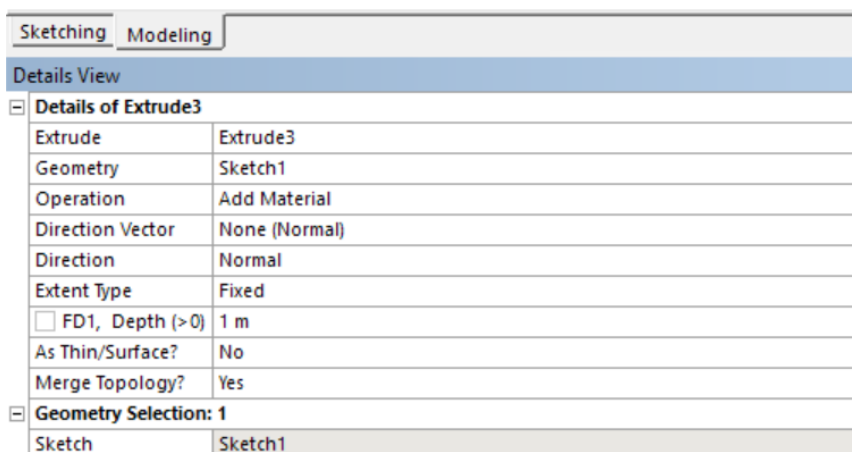
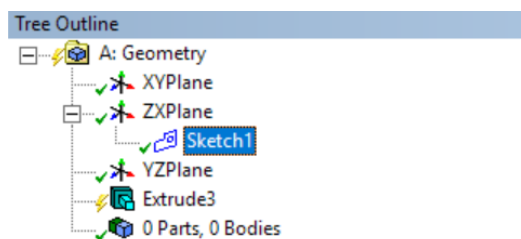
Następnie przechodzimy do **Modeling** – w płaszczyźnie **ZXPlane** pojawi się nasz szkic.



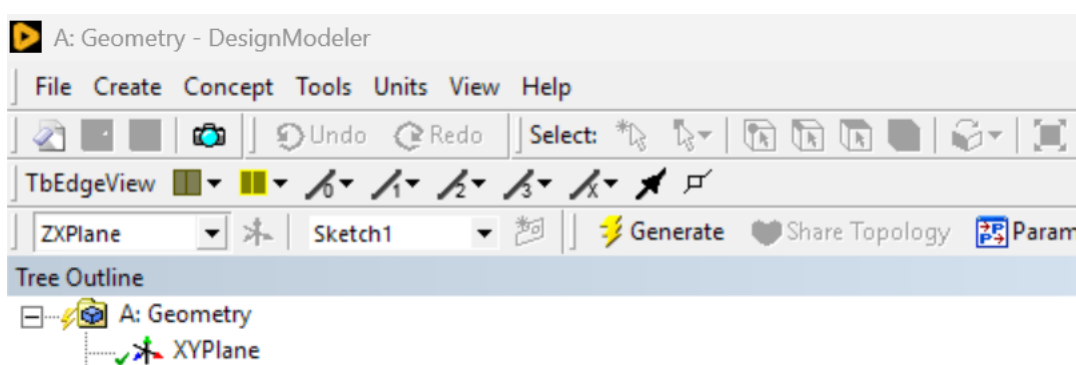
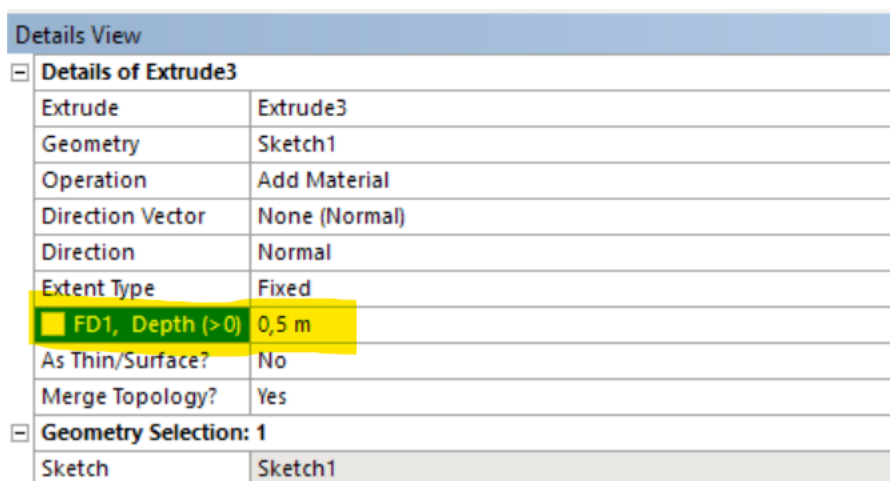
Klikamy Extrude.

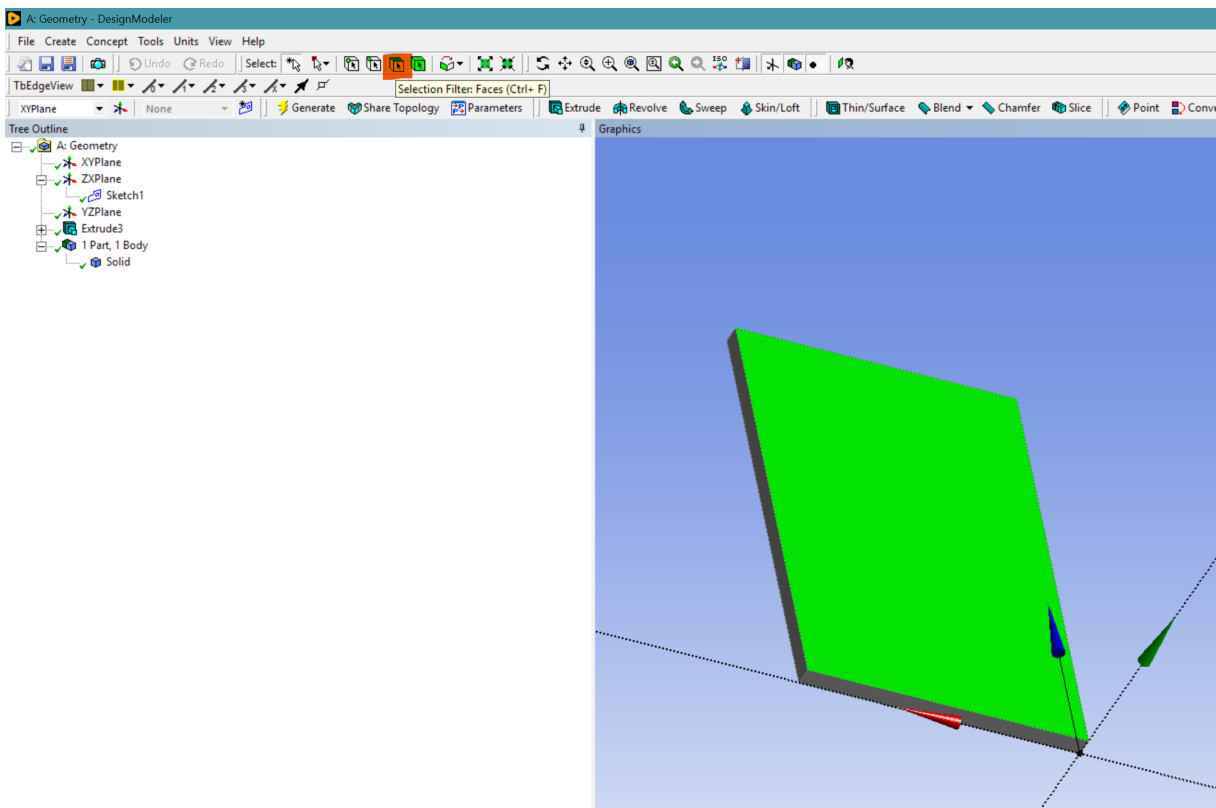
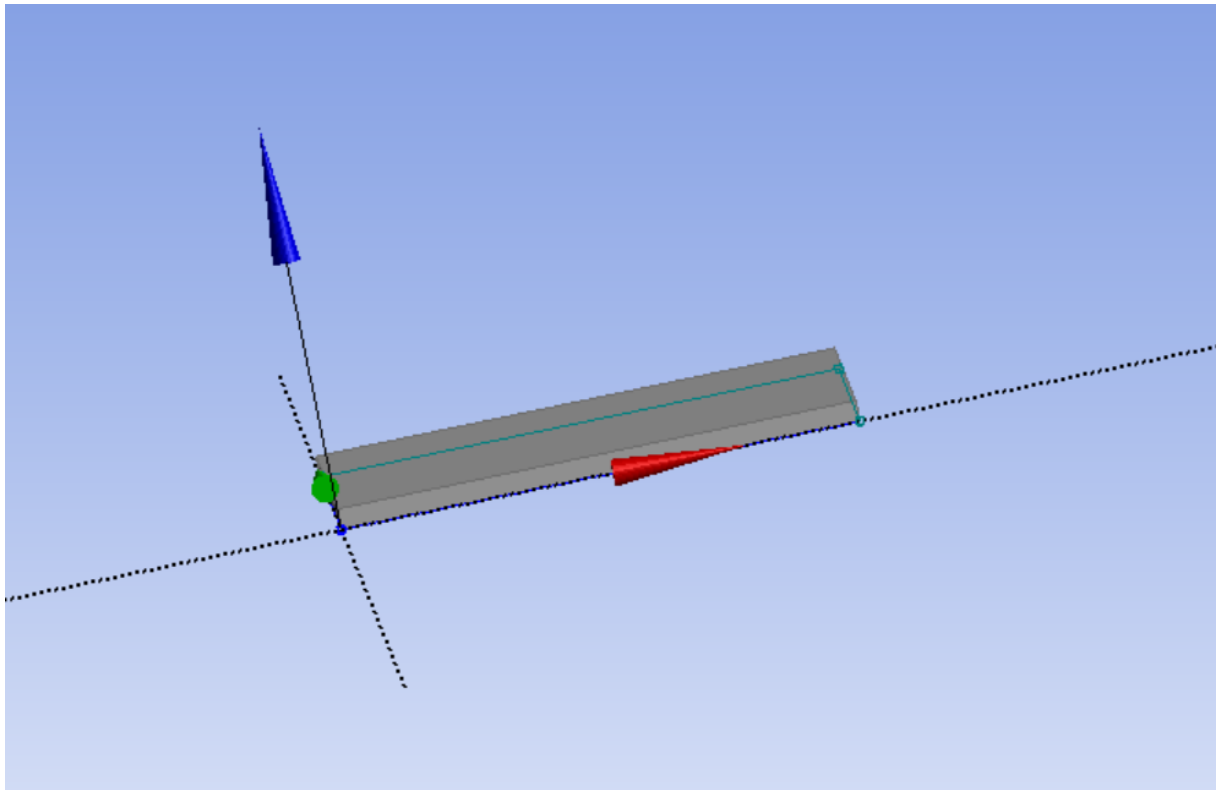


Należy kliknąć **Sketch1**, aby zaznaczyć geometrię, na której działamy i wybrać **Apply**.

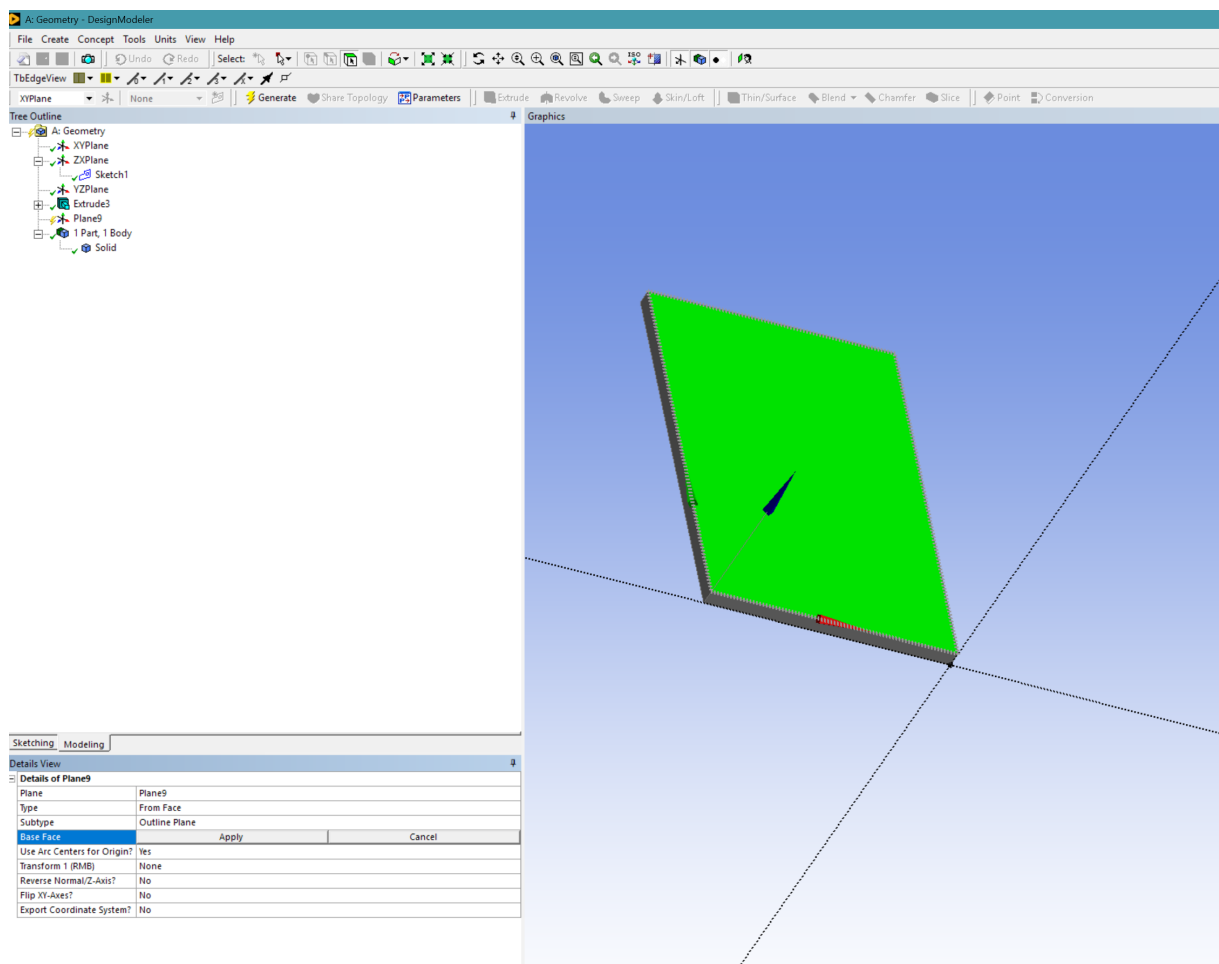
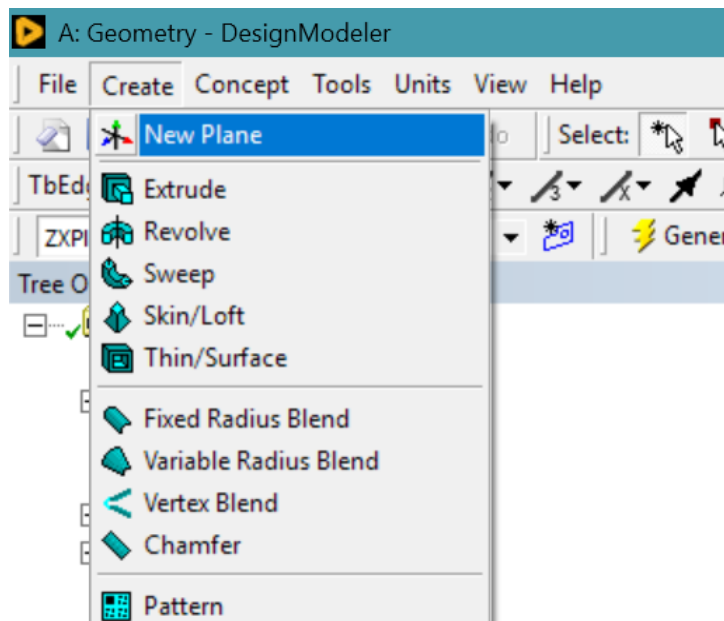


Ustawiamy żadaną wysokość i klikamy **Generate**.





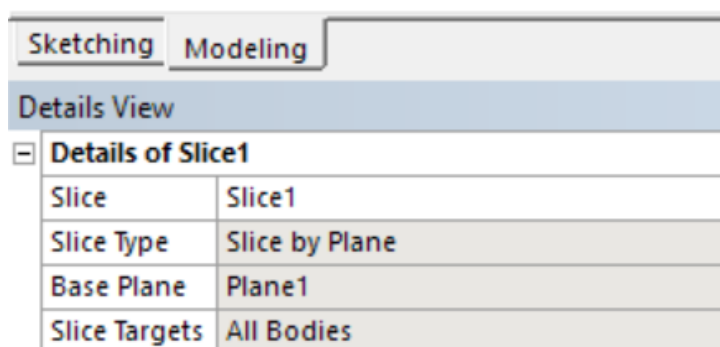
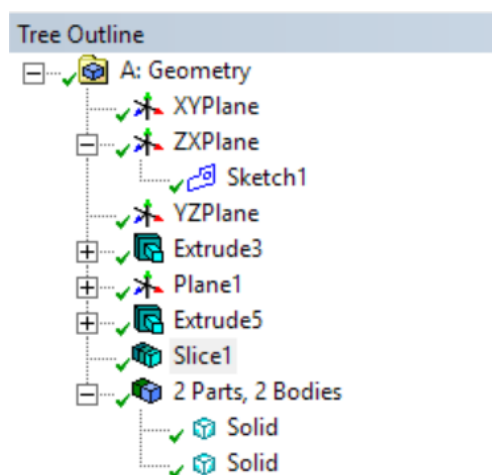
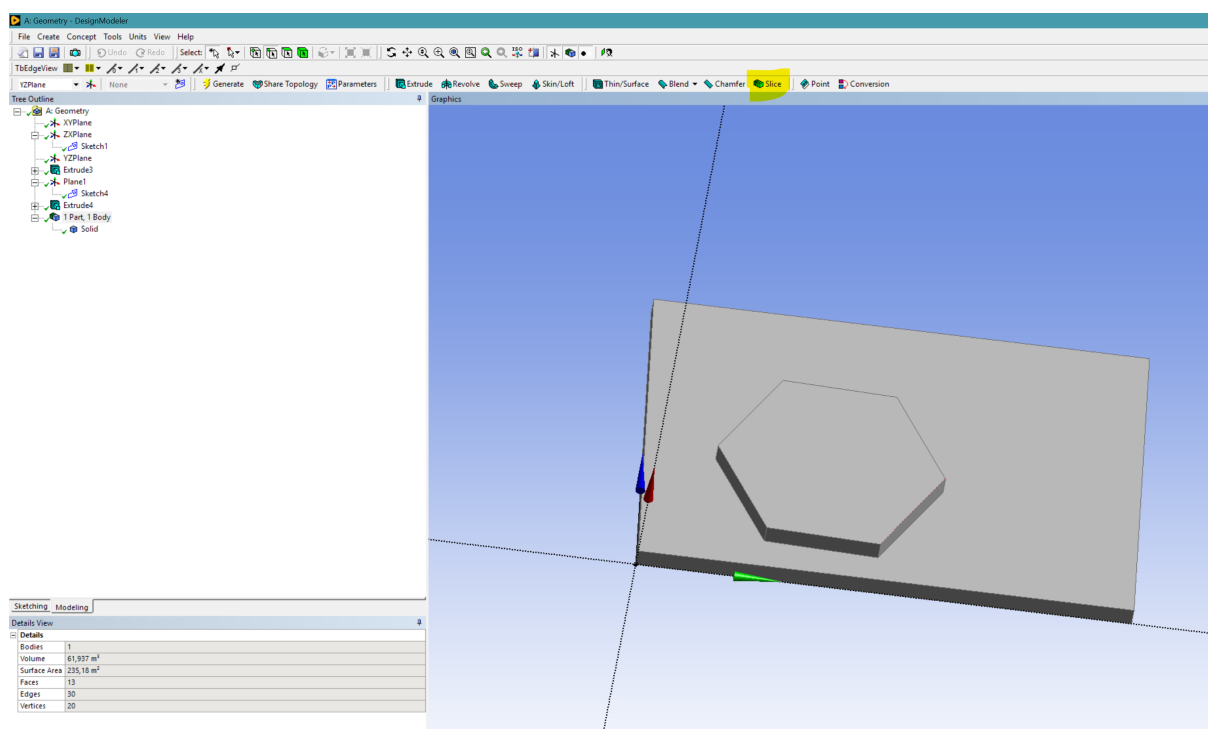
Następnie, by utworzyć nową płaszczyznę **Create** → **New Plane**.



Klikamy **Apply**.

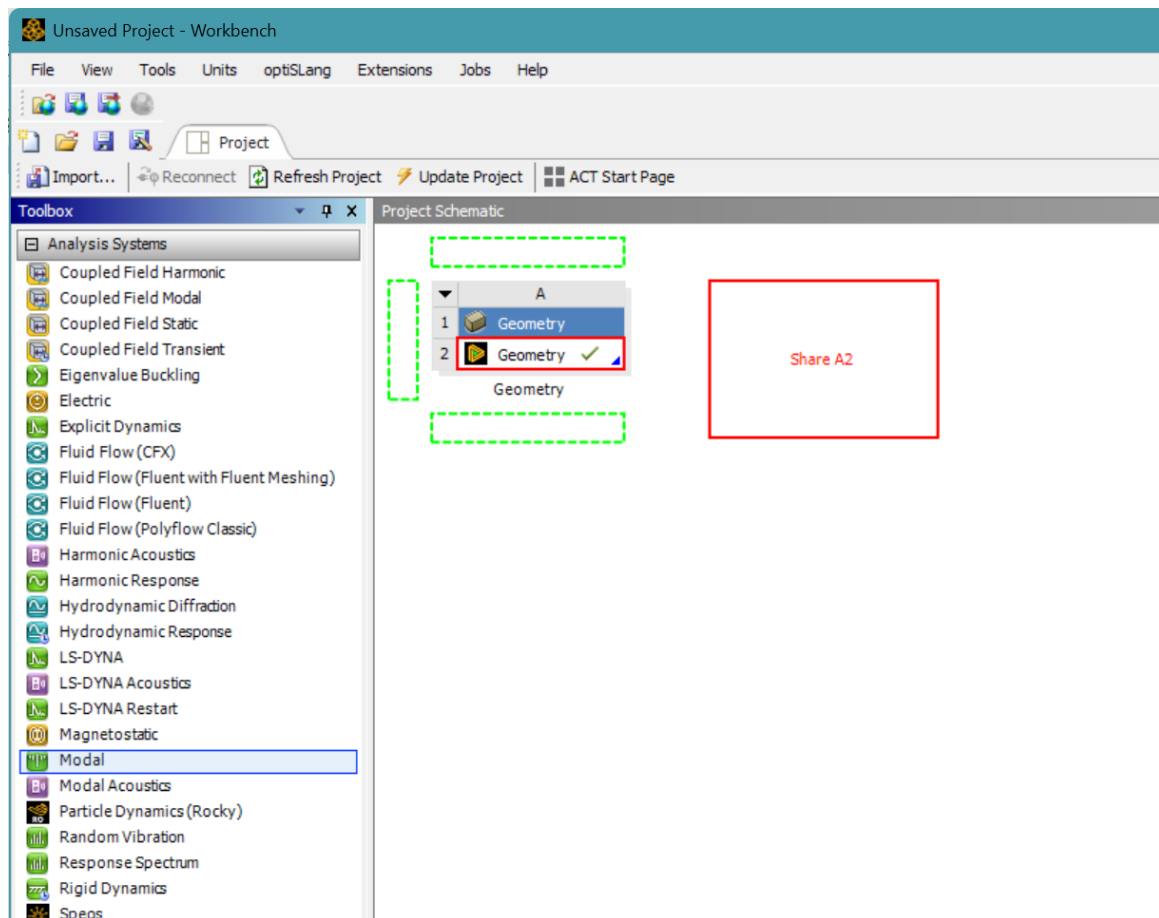
Generujemy kolejne elementy i rysujemy je w ten sam sposób.

Następnie należy dodać **Slice**, aby oddzielić od siebie elementy.

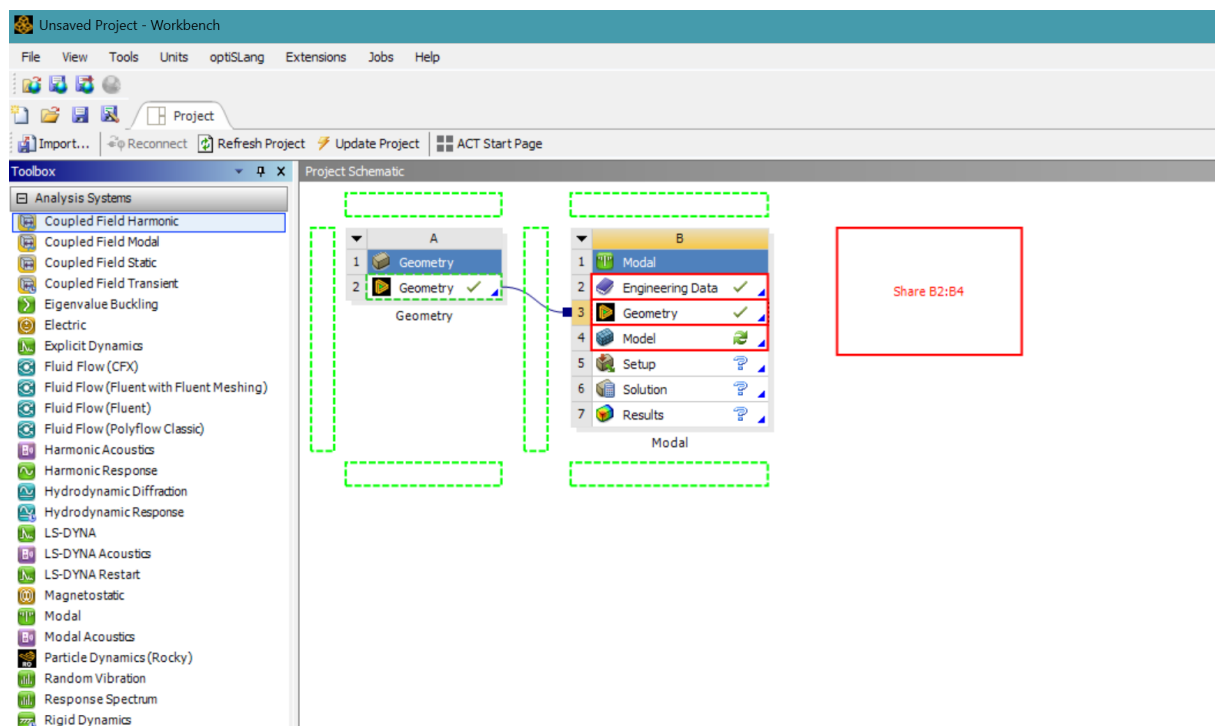


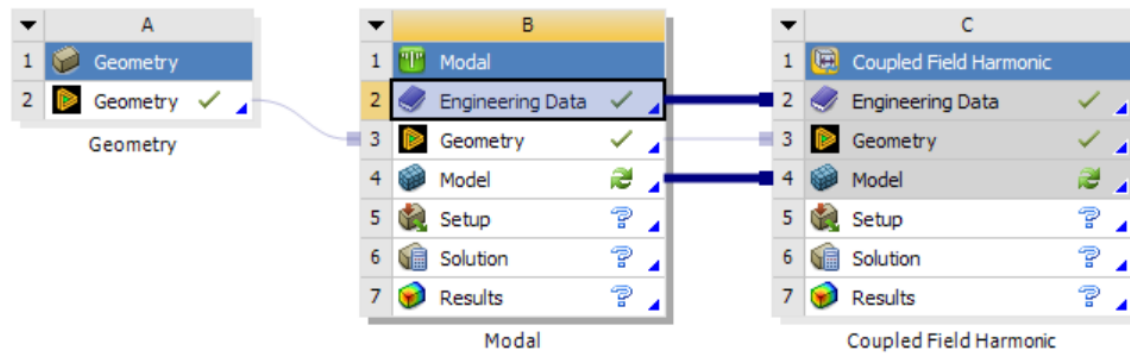
Po zakończeniu zapisujemy zaprojektowaną figurę i przechodzimy do **Workbench**.

Na **Geometry** przeciągamy blok **Model**.

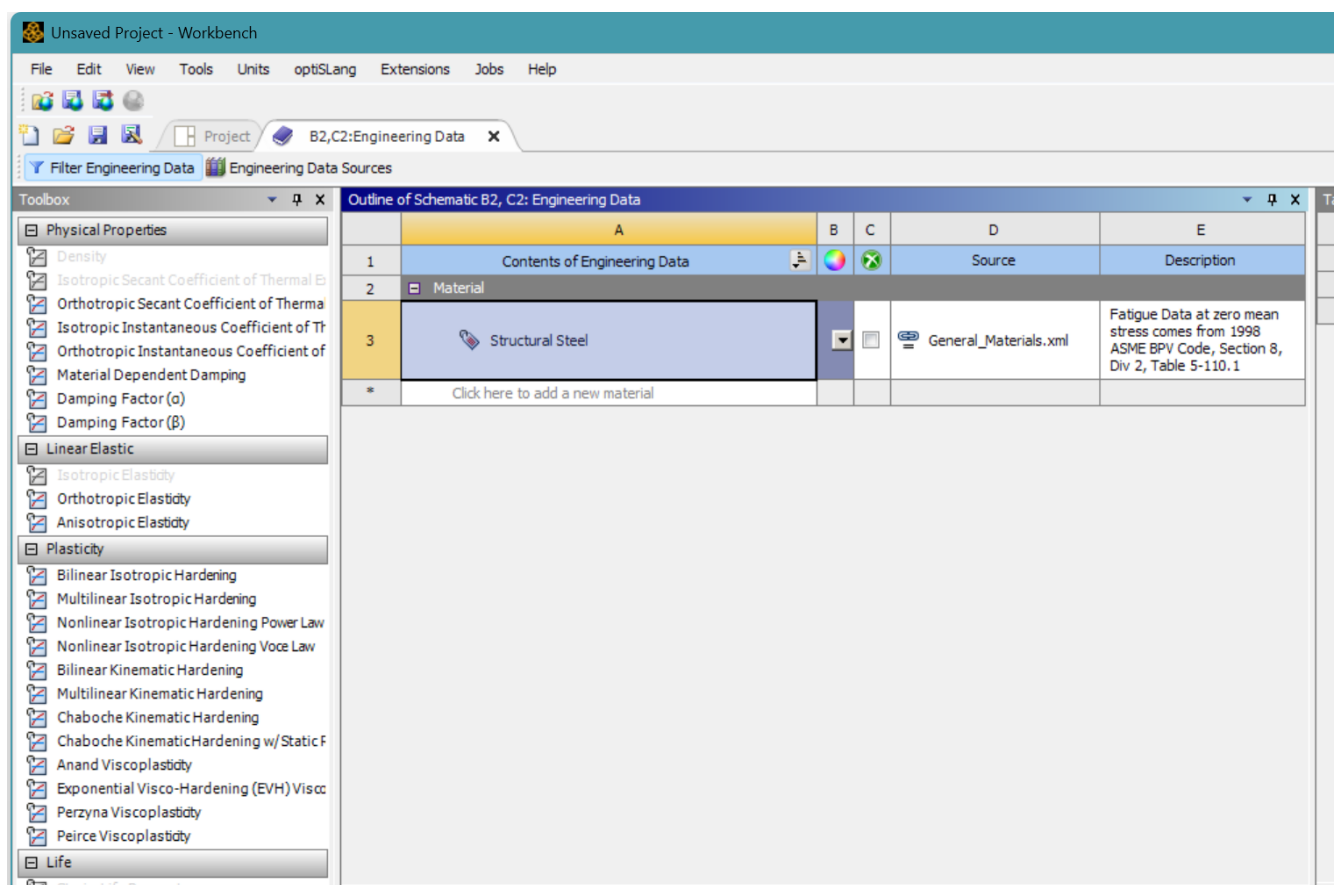


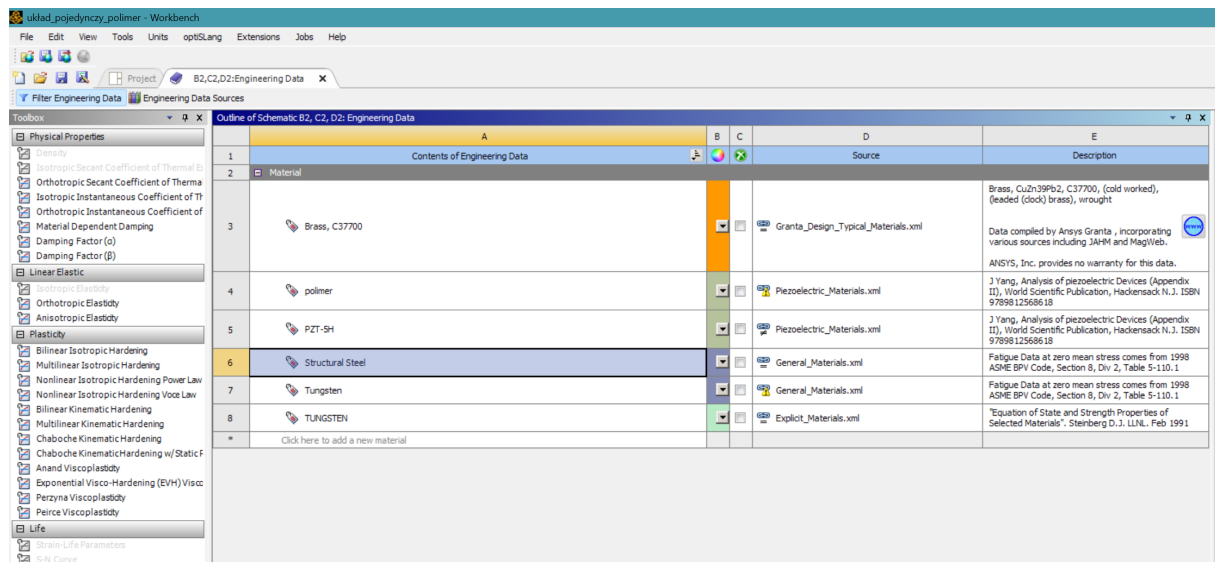
Na **Engineering Data**, **Geometry** oraz **Model** przeciągamy blok **Coupled Field Harmonic**.



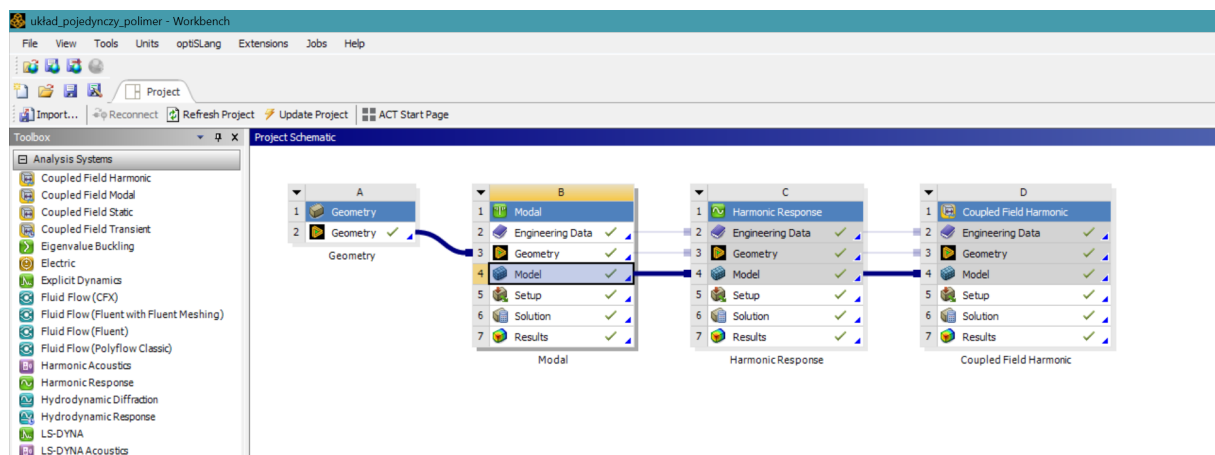


Wchodzimy w **Engineering Data** i dodajemy materiały, z których będziemy chcieli korzystać.

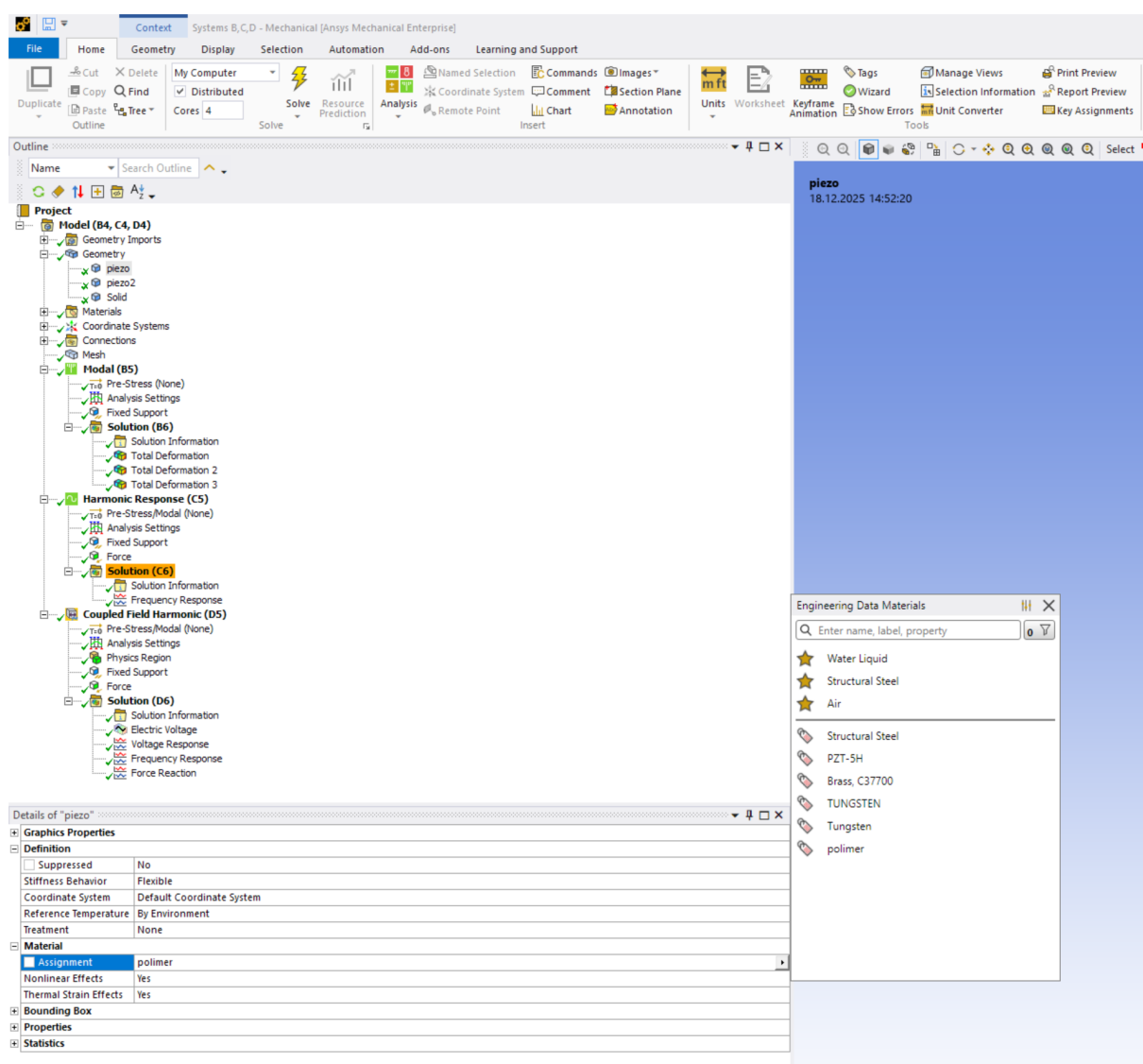




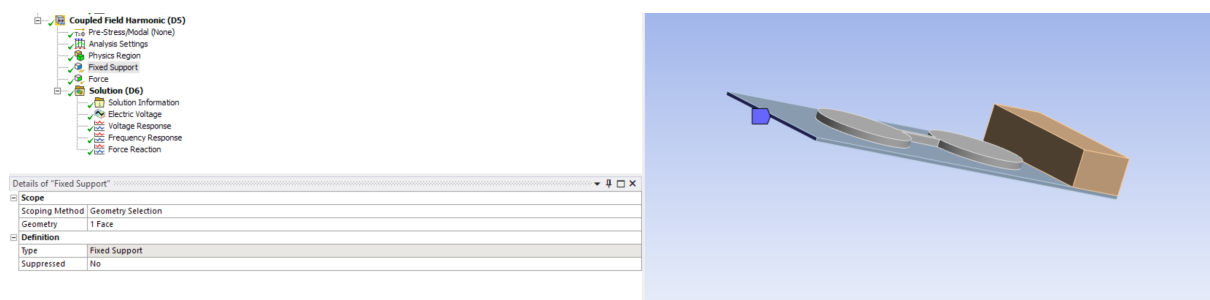
Następnie wracamy do **Workbench** i klikamy dwukrotnie **Model**, aby otworzyć **Mechanical**.

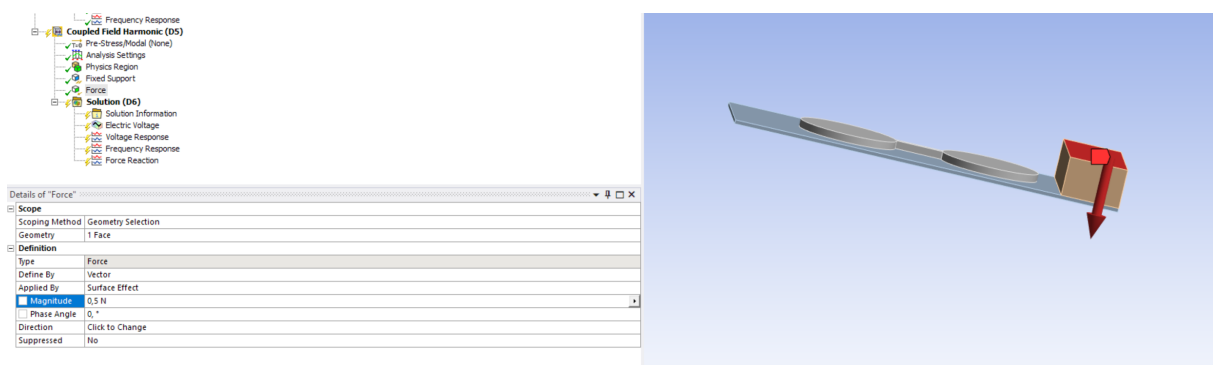


W zakładce **Geometry** klikamy wybrany element i przypisujemy mu odpowiedni **material**.

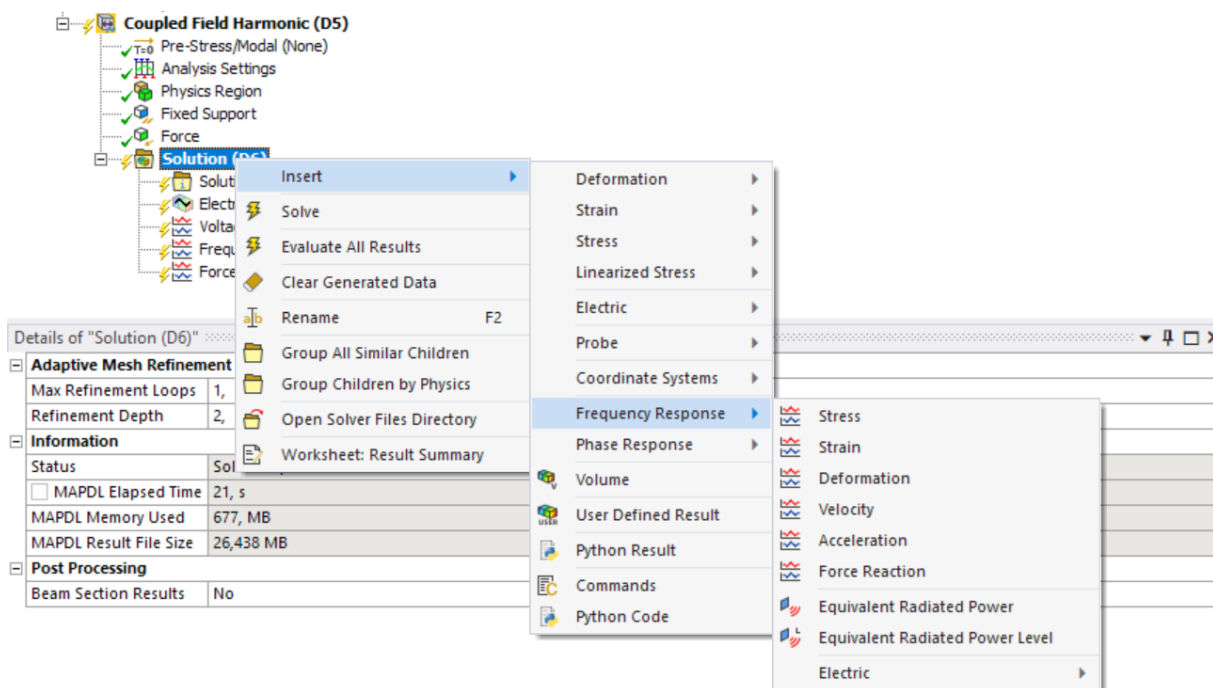


Ustalamy **Fixed Support** oraz dodajemy obciążenie **Force**.





Aby dodać wykresy, przechodzimy do **Solution** → **Insert** → **Frequency Response**.



Na końcu klikamy **Generate** i analizujemy otrzymane wykresy.

The screenshot shows the ANSYS Mechanical Enterprise software interface. The top menu bar includes File, Home, Solution, Display, Selection, Automation, Add-ons, and Learning and Support. The 'Solve' button is highlighted in the 'Solution' tab. A tooltip for the 'Solve (F5)' button reads: "Solve the simulation using the selected solve handler. Press F1 for help." The 'Outline' tree on the left shows the project structure, including Model (B4, C4, D4), Materials, Coordinate Systems, Connections, Mesh, Modal (B5), Solution (B6), Harmonic Response (C5), Solution (C6), and Coupled Field Harmonic (D5). The 'Details of "Solution (D6)"' panel is open, showing the following information:

Details of "Solution (D6)"	
Adaptive Mesh Refinement	
Max Refinement Loops	1
Refinement Depth	2
Information	
Status	Solve Required
MAPDL Elapsed Time	21, s
MAPDL Memory Used	677, MB
MAPDL Result File Size	26,438 MB
Post Processing	
Beam Section Results	No

